COURSE INFORMATION

School/Faculty:	**Faculty of Computing**		
Program name:	**BSc in Computer Science (Computer Networks and Security) with honour** → BSc in CS (Data Eng)		
Course code:	**SECR2033/SCSR1033**	**Academic Session/Semester:**	2025/2026-02
Course name:	**Computer Organization and Architecture**	**Pre/co requisite (course name and code, if applicable):**	Digital Logic
Credit hours:	3		

Course synopsis	This course introduces the principles of computer organization and architecture, focusing on how computer systems operate at a fundamental level. Key topics include system components, computer arithmetic and performance measurement, addressing modes, micro-instructions, and the design and implementation of low-level coding for operational systems. Students will also engage in group-based project deliverables to apply theoretical knowledge in practical contexts. The course employs a combination of lectures, active learning, lab-based work, and project reviews, emphasizing problem-solving, collaboration, and hands-on coding practice. By the end of the course, students will be able to analyze and evaluate computer system performance, differentiate instruction-level operations, design and implement coding solutions, and demonstrate effective teamwork in delivering project-based outcomes.			
Course coordinator (If applicable)	Dr Zuriahati binti Mohd Yunos			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
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	Dr Neo Chin Chea	MJIT		

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

No.	* CLO	PLO ** (MQF Cluster Code)	*** Taxonomies and **** Graduate Attributes	T&L methods	*****Assessment methods
CLO1	Analyze computer system components by applying basic computer arithmetic to evaluate and compare their performance.	PLO1 (C1)	C4 (SC1)	Lecture, Active Learning	Q, T, F
CLO2	Analyze the use of different addressing modes and micro-instructions to determine their effectiveness in computer system operations.	PLO2 (C2)	C2 (SC2)	Lecture, Active Learning	Q, T, F
CLO3	Analyze and troubleshoot low-level coding for operational computer systems through collaborative problem-solving and professional practice.	PLO4 (C3B)	A4 A(SI3)	Project-based learning	L
CLO4	Demonstrate project deliverables in a group based on predefined specifications.	PLO11 (C5)	A3 A(CC3)	Project-based learning	PR, Pr

This is the basic mapping required for the CI. Any added information is allowed (extra columns for weight or other elements) provided this is made consistent for all CI at program/school/faculty level.
* Up to 5 CLO
Refer *** Taxonomies of Learning and **** UTM's Graduate Attributes for UG and Generic Skills for PG, where applicable for measurement of outcomes achievement
***** PS - Problem Solving; T - Test; A - Assignment; Ref - Reflection; PeR - Peer Review; PR - Project; Pr - Presentation; F- Final Examination; AA - Alternative Assessment; THFE - Take Home Final Exam; P - Portfolio, L - Lab; LT - Lab Test; CAD - Case Analysis and Discussion; RP - Role Play; V - Viva Voce; ReP - Research Progress; R - Report.

** MQF Cluster Code
C1 = Knowledge & Understanding, C2 = Cognitive Skills, C3A = Practical Skills, C3B = Interpersonal Skills, C3C= Communication Skills, C3D = Digital Skills, C3E = Numeracy Skills, C3F = Leadership, Autonomy & Responsibility, C4A = Personal Skills, C4B = Enterpreneurial Skills, C5 = Ethics & Professionalism

Details on Innovative T&L practices:

No.	Type	Implementation
1	Active Learning	In-class problem-solving with real datasets.
2	Project-Based Learning	Conducted through a programming assignment. Students, in groups of 3-4, are given 1 project that requires assembly-language solutions, including problem specification and low-level machine code, using Microsoft Visual Studio. Compliance with the problem specification must be documented in written reports and source code. Or CPU performance analysis.

Transferable skills (generic skills learning in course of study which can be useful and utilised in other settings) :
Students' generic skills can be applicable in this course by cultivating teamwork, communication, and professional competence through collaborative coding and system analysis. Students practice accountability, integrity, and ethical decision-making as they evaluate project outcomes. Emphasising patriotism and social responsibility, the course encourages responsible use of computing knowledge to support society and contribute to national technological advancement.

Student learning time (SLT) / Effective Learning Time (ELT) details:

Week/ Meeting	Course Content Outline and Subtopics	CLO*	Learning and Teaching Activities										TOTAL SLT
			Face-to-Face (F2F)								Non F2F Independent Learning		
			Physical				Online (Synchronous)				Online (Asynchronous)	Others	
			L	T	P	O	L	T	P	O			
Week 1	MODULE 1: Basic Concepts and Computer Evolution Course Overview, Main Components of Computer, Computer Structure and Functions, Computer Evolution, Computer Level Hierarchy	1	2	1							2		5
Week 2	MODULE 2: Data Representation in Computer Systems Fixed-Number Representation, Fixed-Number Arithmetic Operations (Addition, Subtraction, Multiplication and Division)	1	2	1							2	1	6
Week 3	… Floating-Number Representation (IEEE 754), Floating-Point Arithmetic (Addition and Multiplication). **Tutorial 1**	3	2	1							2		5
Week 4	MODULE 3: Introduction to Assembly Language Programming Constant, Identifiers, Expression, Data Type, Little Endian, Basic Instructions **Quiz 1 – M1 & M2**	2	3								2	1	6
Week 5	MODULE 3: Introduction to Assembly Language Programming Constant, Identifiers, Expression, Data Type, Little Endian, Basic Instructions **Lab installation software (VS and MASM) and Lab 1**	2	2	1							2		5

Week 6	MODULE 4: Instruction Set Architecture (ISA) ISA Level, Element of Instructions, Instruction Types, Number of Addresses	2	3								3		6
Week 7	MODULE 4: Instruction Set Architecture (ISA) … Registers, Type of Operands, Addressing Modes, Instruction Format **Quiz 2 – M3&4**	3	3								2		5
Week 8	**SEM BREAK (4 – 10 May 2026)**												
Week 9	MODULE 5: Central Processing Unit (CPU) Datapath and Control Unit, Bus, Registers, Instruction Cycle (Fetch, Indirect, Execution, Interrupt) **Lab 2 (xchg, dec, inc, neg)**	2	3								2	1	6
Week 10	Pipeline, Analogy Pipeline, Degree of Speedup, Instruction Pipeline, Pipeline Limitations, Resource Conflict, Data Dependency, Branch Difficulties **Lab 3 (flags, offset, arrays, jmp, loop)** **Mid-Term (M2 - M4)** **(20/5/2026 - Wednesday Night)**	4	3								2		5
Week 11	Control Unit (CU) Operation, Micro-operations, CU Functions, Microinstruction, Micro-programmed CU, Control Memory **Tutorial 2** **Final Project Activity**	3	2	1							3		6
Week 12	MODULE 8: Performance Measurements and Analysis Computer Performance Measures and Equation, CPU Execution Time, CPI, Increasing CPU Performance, Comparing CPU Performance, Average System Performance, Comparing Relative Performance, Benchmarking **Tutorial 3**	1	2	1							2	1	6
Week 13	MODULE 6: Memory System Memory Hierarchy, Random Access, Direct Access and Sequential Access, Main Memory - Types; RAM and ROM, Memory Location and Capacity, Latency and Cycle Time, Transfer Rate and Bandwidth, Memory Interleaving, HOI and LOI **Quiz 3 – M5**	4	2	1							2		5

14	Cache Memory: Cache Memory Principle, Cache Address Mapping Schemes - Direct Mapping, Block Direct Mapping, Fully Associative Mapping, Set Associative Mapping, Replacement Policy, Write Policy, Average Memory Access Time, Multilevel Caches, Virtual Memory **Tutorial 4**		2	1							2	1	6
15	MODULE 7: Input / Output (I/O) and Storage System I/O and Performance, I/O Architectures - Programmed I/O, Interrupt Driven I/O, Direct Memory Access (DMA), Channel I/O, Storage Systems, Magnetic Disk Technology, Optical Disks, Magnetic Tape, RAID		2	1							2		5
16	**Revision Week**												
											SUB-TOTAL SLT :		**77**

Continuous Assessment		**%**	**Face-to-Face (F2F)**		**Non F2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	Assignment (Lab 1 - 3)	15	6		2	6	14
2	Quiz (1 - 3)	15	2			2	4
3	Tutorial (1 – 4)	10	2			2	4
4	Mid-Term Test	15				4	4
5	Group Project Report & Presentation	15				5	5
					SUB-TOTAL SLT :		**31**

Summative Assessment		**%**	**Face-to-Face (F2F)**		**Non F2F Independent Learning for Assessment**		**TOTAL SLT**
			Physical	**Online (Synchronous)**	**Online (Asynchronous)**	**Others**	
1	(F) Final Exam	30	3			9	12
					SUB-TOTAL SLT :		**12**
					SLT for Assessment :		**43**
					GRAND TOTAL SLT :		**120**

A	% SLT for F2F Physical Component :	52.025
B	% SLT for Online & Independent Learning Component :	56.075
C	% SLT for Online Component :	32.00
D	% SLT for All Practical Component :	0.00
D1	% SLT for F2F Physical Practical Component :	0.00
D2	% SLT for F2F Online Practical Component :	0.00
Please tick (✓) if this course is Industrial Training/ Clinical Placement/ Practicum using 50% of Effective Learning Time (ELT)		

Identify special requirement or resources to deliver the course (e.g., software, nursery, computer lab, simulation room etc)
Computer lab with MS Visual Studio 2019

References (include required and further readings, and should be the most current)

Text book
Patricio Bulic, Understanding Computer Organization. Springer, 2024
W. Stalling, Computer Organization and Architecture, 10th Edition, Prentice Hall, 2016.
L. Null & J. Lobur, The Essentials of Computer Organization and Architecture, 4th Edition, 2015. Kip R. Irvine, Assembly Language for Intel-based Computers, Prentice Hall, 6th edition, 2011.

Additional references
David Patterson and John Hennessy, Computer Organization and Design: The Hardware/Software Interface, 5th Edition, Morgan Kaufmann, 2014. Mano, M. Morris, Computer System Architecture, 3rd Edition, Prentice-Hall, 1993.

Academic honesty and plagiarism: (Below is just a sample)

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES)
Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of zero for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (if applicable)

All students must follow the passing requirements of a computing course: obtain at least the minimum passing marks as specified by UTM regulations (40% or a grade of D+).
For **BSE Programme** passing mark (SCSR1033):
- Pass both the continuous assessment AND the final assessment, AND
- Get at least the minimum pass marks following UTM regulation (40%).
- The passing mark for Continuous Assessment (PB) is 30% of coursework marks.
- The passing mark for the Final Assessment (PA) is 20% of the final exam marks.

No	Assessment	PLO1 (KW)		PLO3 (PS)	PLO8 (LAR)	PLO9 (PRS)	TOTAL
		CLO1	CLO2	CLO3	CLO4	CLO4	
1	LABS (3)			15			15
2	QUIZ (3)	8	7				15
3	Tutorial (4)	5	5				10
4	MID-TERM	5	10				15
5	FINAL EXAM	20	10				30
6	PROJECT REPORT				10	5	15
	TOTAL PLO	70		15	10	5	100

subject to chg

Disclaimer:

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While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.

ELT = (Theory + Industrial Guidance + Assessment) x 50%
Total of credit for LI/Practical = ELT/40 Notional Hours

Note: For ODL Programme : Courses with mandatory practical requirement imposed by programme standards or any related standards can be exempted from complying to the minimum 80% ODL delivery rule in the SLT.

Prepared by :		**Certified by :**	
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Signature :		**Signature :**	
Date :		**Date :**	